

Cloud computing is **the on-demand delivery of IT resources—such as servers, storage, databases, and software—over the internet with pay-as-you-go pricing, rather than owning and maintaining physical data centers**. It enables companies to lower costs, scale infrastructure instantly, and access services from providers like AWS, Azure, and Google Cloud.

Key Aspects of Cloud Computing

- **Service Models:**
 - **IaaS (Infrastructure as a Service):** Provides fundamental computing, network, and storage resources.
 - **PaaS (Platform as a Service):** Offers a development environment to build and deploy apps.
 - **SaaS (Software as a Service):** Delivers software applications over the internet (e.g., Salesforce, Office 365).
- **Deployment Models:**
 - **Public Cloud:** Services delivered over the internet to the public, like [AWS](#) or [Google Cloud](#).
 - **Private Cloud:** Dedicated infrastructure for a single organization.
 - **Hybrid Cloud:** A combination of public and private clouds.
- **Core Benefits:**
 - **Cost Efficiency:** Shifts capital expenses (CapEx) to operating expenses (OpEx).
 - **Speed & Agility:** Resources can be provisioned in minutes.
 - **Global Scale:** Ability to expand to different regions instantly.
 - **Reliability:** [Data backup](#) and disaster recovery are simplified.

Common Examples

- **Storage:** Google Drive, Dropbox.
- **Streaming:** Netflix, Spotify.
- **Productivity:** Gmail, Slack.
- **Infrastructure:** AWS, Azure, Google Cloud.